

Technical University of Denmark

Written examination: 24 May 2018, 9 AM - 1 PM.

Course name: Introduction to Machine Learning and Data Mining.

Course number: 02450.

Aids allowed: All aids permitted.

Exam duration: 4 hours.

Weighting: The individual questions are weighted equally.

Your answers to the questions are to be handed in using the electronic file. Use only this page for hand in if you are unable to hand in digitally. In case you have to hand in the answers using the form on this sheet, please print your name and student number clearly.

The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. Correct answer gives 3 points, wrong answer gives -1 point, and “Don’t know” (E) gives 0 points.

The individual questions are answered by filling in the answer fields with one of the letters A, B, C, D, or E.

Answers:

1	2	3	4	5	6	7	8	9	10
C	A	B	B	D	A	C	D	A	C
11	12	13	14	15	16	17	18	19	20
B	B	D	A	B	D	C	C	B	C
21	22	23	24	25	26	27			
C	A	B	D	A	D	C			

Name: _____

Student number: _____

PLEASE HAND IN YOUR ANSWERS DIGITALLY.

**USE ONLY THIS PAGE FOR HAND IN IF YOU ARE
UNABLE TO HAND IN DIGITALLY.**

No.	Attribute description	Abbrev.
x_1	Number of seats times kilometers pr. week	S*KM/Week
x_2	Incidences 1985-1999	Inc. 85-99
x_3	Fatal accidents 1985-1999	FA 85-99
x_4	Fatalities 1985-1999	Fat. 85-99
x_5	Incidences 2000-2014	Inc. 00-14
x_6	Fatal accidents 2000-2014	FA 00-14
y	Fatalities 2000-2014	Fat. 00-14

Table 1: The attributes of the airline safety dataset that contains 56 observations of different airline companies and their properties in terms of number of seats times number of kilometers per week, incidences, fatal accidents, and fatalities accumulated over the period of 1985-1999 and 2000-2014 respectively. We presently consider as output y the number of fatalities from 2000-2014.

Question 1. We will consider the airline safety dataset consisting of 56 airline companies and their number of flights as quantified by number of seats times kilometers per week as well as incidences, fatal accidents, and fatalities quantified for the period of 1985-1999 and 2000-2014 respectively¹. For brevity this dataset will be denoted the airline safety dataset. In Table 1 is given the attributes of the data as well as the output attribute y defined by the number of fatalities from 2000-2014. In Figure 1 is shown a matrix plot of the six attributes x_1-x_6 .

Considering the attributes described in Table 1 and the matrix plot in Figure 1 which one of the following statements regarding the attributes x_1-x_6 is correct?

- A. At least one of the attributes appears to be normal distributed.
- B. x_2 corresponding to Inc. 85-99 and x_3 corresponding to FA 85-99 are negatively correlated.
- C. All the six attributes are ratio.**
- D. All the attributes are continuous.
- E. Don't know.

Solution 1. Inspecting the histograms along the diagonal of the matrix plot it is clearly seen that

¹The dataset is taken from <https://github.com/fivethirtyeight/data/tree/master/airline-safety>

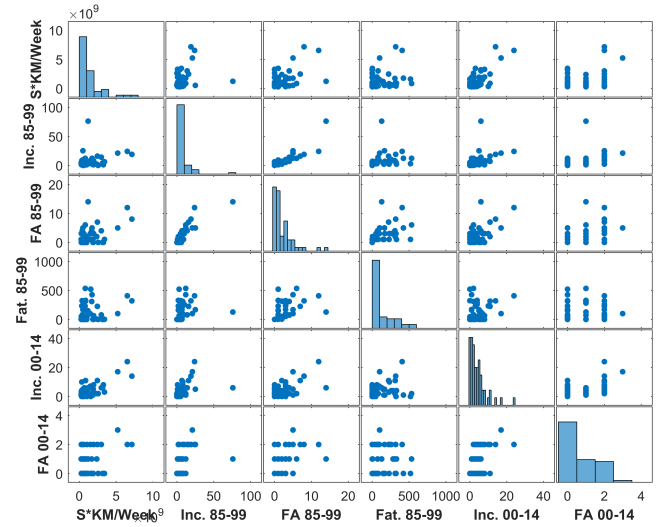


Figure 1: Matrix plot of the six attributes x_1-x_6 . Along the diagonal of the matrix plot is given the histogram of each attribute.

they all have a mode around zero/low values and decrease with no negative observations on the left side of the mode. This is not corresponding to the bell shape of a normal distribution and thus none of the attributes appear normally distributed. Inspecting, the plot of x_2 vs. x_3 we observe that airline with more incidences in 85-99 also have more fatal accidents in 85-99, there is thus a positive correlation between these two attributes. As for all attributes zero means absence of what is being measured and it makes sense to talk about a company having twice as many seats times kilometers per week, or incidents, or accidents, or fatalities, thus, all these attributes are ratio. As incidents, accidents and fatalities are counts they are discrete integer variables and not continuous.

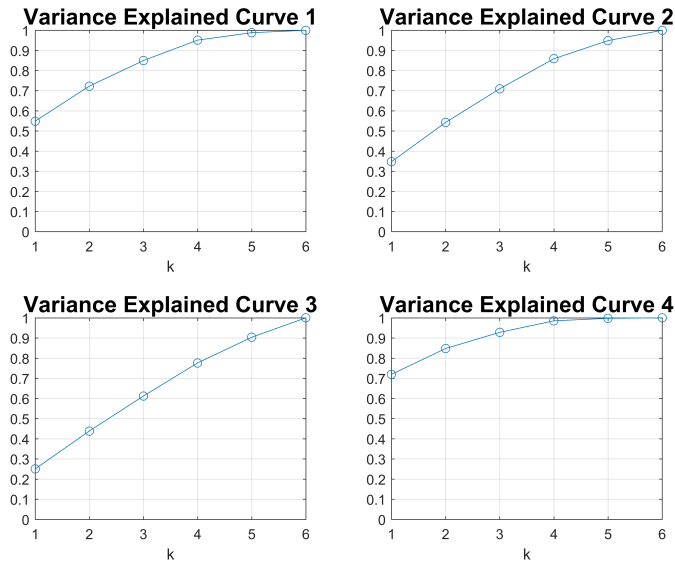


Figure 2: Four different curves of the variance explained as function of keeping the first k principal components when performing a PCA analysis. One of the four curves correctly corresponds to the PCA of the standardized airline safety data.

Question 2. A principal component analysis (PCA) is carried out on the standardized attributes x_1-x_6 , forming the standardized matrix $\tilde{\mathbf{X}}$ (i.e., each attribute has been subtracted its mean and divided by its standard deviation). A singular value decomposition is applied to the standardized data matrix, i.e. $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$ and we find the following solution in terms of the $\mathbf{S}$ and $\mathbf{V}$ matrices:

$$\mathbf{S} = \begin{bmatrix} 13.5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7.6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5.8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3.5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2.0 \end{bmatrix}$$

$$\mathbf{V} = \begin{bmatrix} 0.38 & -0.51 & 0.23 & 0.47 & -0.55 & 0.11 \\ 0.41 & 0.41 & -0.53 & 0.24 & 0.00 & 0.58 \\ 0.50 & 0.34 & -0.13 & 0.15 & -0.05 & -0.77 \\ 0.29 & 0.48 & 0.78 & -0.17 & 0.00 & 0.23 \\ 0.45 & -0.42 & 0.09 & 0.03 & 0.78 & 0.04 \\ 0.39 & -0.23 & -0.20 & -0.82 & -0.30 & 0.04 \end{bmatrix}$$

In Figure 2 is given the pct. of variance explained by retaining the first k principal components as a function of k . Which one of the four curves corresponds to the correct curve of variance explained as function of the number of principal components retained?

- A. Variance Explained Curve 1.
- B. Variance Explained Curve 2.
- C. Variance Explained Curve 3.
- D. Variance Explained Curve 4.
- E. Don't know.

Solution 2. The variance explained by the first k principal components is given by $\frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^6 \sigma_i^2}$. We thereby get that the curve should have the following values:

$$k=1: \frac{13.5^2}{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2+2.0^2} = 0.5487$$

$$k=2: \frac{13.5^2+7.6^2}{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2+2.0^2} = 0.7226$$

$$k=3: \frac{13.5^2+7.6^2+6.5^2}{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2+2.0^2} = 0.8498$$

$$k=4: \frac{13.5^2+7.6^2+6.5^2+5.8^2}{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2+2.0^2} = 0.9511$$

$$k=5: \frac{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2}{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2+2.0^2} = 0.9880$$

$$k=6: \frac{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2+2.0^2}{13.5^2+7.6^2+6.5^2+5.8^2+3.5^2+2.0^2} = 1$$

Only Variance Explained Curve 1 has this property.

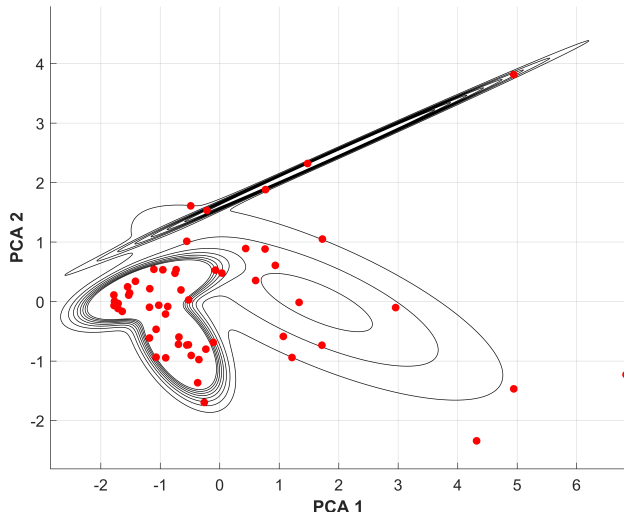


Figure 3: A Gaussian Mixture Model (GMM) using four clusters fitted to the standardized airline safety data projected onto the first two principal components.

Question 3. According to the extracted PCA directions given by the matrix $\mathbf{V}$ in the above what will be the coordinate of the standardized observation $\tilde{\mathbf{x}}^* = [-0.1 \ 0.2 \ 0.1 \ -0.3 \ 1 \ 0.5]$ when projected onto the first two principal components?

- A. (-0.753, 0.206)
- B. (0.652, -0.512)**
- C. (0.680, 0.019)
- D. (0.671, -0.139)
- E. Don't know.

Solution 3. The observation $\tilde{\mathbf{x}}^* = [-0.1 \ 0.2 \ 0.1 \ -0.3 \ 1 \ 0.5]$ will have the projection onto the two first principal components given by

$$[-0.1 \ 0.2 \ 0.1 \ -0.3 \ 1 \ 0.5] \begin{bmatrix} 0.38 & -0.51 \\ 0.41 & 0.41 \\ 0.50 & 0.34 \\ 0.29 & 0.48 \\ 0.45 & -0.42 \\ 0.39 & -0.23 \end{bmatrix} = [0.652 \ -0.512].$$

Thus, the observation will in the projection be located at (0.652, -0.512).

Question 4. Which one of the following statements regarding Gaussian Mixture Modeling (GMM) is correct?

- A. The number of clusters used in the GMM can be determined by selecting the number of clusters that provides the best likelihood of the training data used for training the density.
- B. For high-dimensional data, i.e. where the number of features M is large, it can be beneficial to constrain the covariance of each cluster to be diagonal, i.e. enforcing off-diagonal terms of the covariance matrices to be zero, in order to reduce the number of parameters in the GMM model.**
- C. The GMM is guaranteed to find the optimal clustering for a given dataset.
- D. Similar to the k-means algorithm that assigns observations to the cluster in closest proximity, the EM-algorithm used to estimate the parameters of the GMM considers only the cluster each observation is the most likely to belong to when estimating the parameters in the M-step.
- E. Don't know.

Solution 4. For the GMM we can use cross-validation to determine the number of clusters, however, this selection of the number of clusters must be based on the likelihood of the test data and not the training data, as this otherwise would result in overfitting of the density to the data. For high-dimensional data the covariance matrix can be ill-determined and it can therefore be beneficial to reduce the covariance matrix to a diagonal matrix for which only the variance of each features need to be estimated, which substantially reduces the number of free parameters in the GMM. The GMM is prone to local minima solutions and therefore it is recommended to fit the model using many restarts taking the best of these randomly initialized models. In the E-step of the GMM it is quantified how likely it is for the observations to belong to each cluster and thereby the observations are “soft” assigned to each cluster. Thus, in the subsequent M-step this soft-assignment is used taking into account how likely it is for the observations to belong to each of the clusters and contributing in the update of the cluster according to this, such that each clusters mean and covariance is a weighted average of the observations contribution weighted according to this probability.

Question 5. We fit a Gaussian Mixture Model (GMM) to the standardized data projected onto the first two principal component directions using four mixture components (i.e., 4 clusters). We recall that the multivariate Gaussian distribution is given by:

$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{M/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$, with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Which one of the following GMM densities corresponds to the fitted density given in Figure 3?

A.

$$\begin{aligned} p(\mathbf{x}) = & 0.0673 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.8422 \\ 2.4306 \end{bmatrix}, \begin{bmatrix} 0.2639 & 0.0803 \\ 0.0803 & 0.0615 \end{bmatrix}\right) \\ & + 0.3360 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -1.2222 \\ 0.1830 \end{bmatrix}, \begin{bmatrix} 3.8237 & 1.7104 \\ 1.7104 & 0.7672 \end{bmatrix}\right) \\ & + 0.2992 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -0.6687 \\ -0.7343 \end{bmatrix}, \begin{bmatrix} 0.1166 & -0.0771 \\ -0.0771 & 0.1729 \end{bmatrix}\right) \\ & + 0.2975 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.6359 \\ -0.0183 \end{bmatrix}, \begin{bmatrix} 4.0475 & -1.5818 \\ -1.5818 & 1.1146 \end{bmatrix}\right) \end{aligned}$$

B.

$$\begin{aligned} p(\mathbf{x}) = & 0.0673 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.8422 \\ 2.4306 \end{bmatrix}, \begin{bmatrix} 3.8237 & 1.7104 \\ 1.7104 & 0.7672 \end{bmatrix}\right) \\ & + 0.3360 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -1.2222 \\ 0.1830 \end{bmatrix}, \begin{bmatrix} 0.2639 & 0.0803 \\ 0.0803 & 0.0615 \end{bmatrix}\right) \\ & + 0.2992 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -0.6687 \\ -0.7343 \end{bmatrix}, \begin{bmatrix} 4.0475 & -1.5818 \\ -1.5818 & 1.1146 \end{bmatrix}\right) \\ & + 0.2975 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.6359 \\ -0.0183 \end{bmatrix}, \begin{bmatrix} 0.1166 & -0.0771 \\ -0.0771 & 0.1729 \end{bmatrix}\right) \end{aligned}$$

C.

$$\begin{aligned} p(\mathbf{x}) = & 0.2975 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.8422 \\ 2.4306 \end{bmatrix}, \begin{bmatrix} 3.8237 & 1.7104 \\ 1.7104 & 0.7672 \end{bmatrix}\right) \\ & + 0.3360 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -1.2222 \\ 0.1830 \end{bmatrix}, \begin{bmatrix} 0.2639 & 0.0803 \\ 0.0803 & 0.0615 \end{bmatrix}\right) \\ & + 0.2992 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -0.6687 \\ -0.7343 \end{bmatrix}, \begin{bmatrix} 0.1166 & -0.0771 \\ -0.0771 & 0.1729 \end{bmatrix}\right) \\ & + 0.0673 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.6359 \\ -0.0183 \end{bmatrix}, \begin{bmatrix} 4.0475 & -1.5818 \\ -1.5818 & 1.1146 \end{bmatrix}\right) \end{aligned}$$

D.

$$\begin{aligned} p(\mathbf{x}) = & 0.0673 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.8422 \\ 2.4306 \end{bmatrix}, \begin{bmatrix} 3.8237 & 1.7104 \\ 1.7104 & 0.7672 \end{bmatrix}\right) \\ & + 0.3360 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -1.2222 \\ 0.1830 \end{bmatrix}, \begin{bmatrix} 0.2639 & 0.0803 \\ 0.0803 & 0.0615 \end{bmatrix}\right) \\ & + 0.2992 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} -0.6687 \\ -0.7343 \end{bmatrix}, \begin{bmatrix} 0.1166 & -0.0771 \\ -0.0771 & 0.1729 \end{bmatrix}\right) \\ & + 0.2975 \cdot \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix} 1.6359 \\ -0.0183 \end{bmatrix}, \begin{bmatrix} 4.0475 & -1.5818 \\ -1.5818 & 1.1146 \end{bmatrix}\right) \end{aligned}$$

E. Don't know.

Solution 5. Inspecting the GMM density we observe that the cluster located at $\begin{bmatrix} 1.8422 \\ 2.4306 \end{bmatrix}$ will have the lowest mixing proportion as only few observations belong to this cluster. Furthermore, it must have a large positive correlation and variance as given by $\begin{bmatrix} 3.8237 & 1.7104 \\ 1.7104 & 0.7672 \end{bmatrix}$. Only answer option 2 and 4 have this property. The cluster located at $\begin{bmatrix} 1.6359 \\ -0.0183 \end{bmatrix}$ has negative covariance but the variance of the cluster is also much larger than the variance of the other cluster having negative covariance. Thus, this cluster must have the covariance $\begin{bmatrix} 4.0475 & -1.5818 \\ -1.5818 & 1.1146 \end{bmatrix}$. Only answer option 4 has this property.

Question 6. We would like to predict the safety of a given airline company. However, in order to take into account the volume of flights of the company when evaluating its safety we define a new output variable given by $\tilde{y} = y/x_1$. By defining $\tilde{y}$ as the number of fatalities divided by the number of seats times kilometers per week of the company, fatalities are quantified relative to the volume of flights that has been catered by the company. A least squares linear regression model is trained using different combinations of the five attributes x_2, x_3, x_4, x_5 , and x_6 in order to predict $\tilde{y}$. Table 2 provides the training and test root-mean-square error (RMSE= $\sqrt{\frac{1}{N} \sum_{i=1}^N (\tilde{y}_i - \hat{y}_i)^2}$) performance of the least squares linear regression model when trained using different combinations of the five attributes. Which one of the following statements is correct?

- A. Forward and backward selection will result in the same features being selected.**
- B. Forward selection will terminate with four features in the feature set.
- C. Backward selection will not remove any features.
- D. x_4 will be among the features selected by forward selection.
- E. Don't know.

Solution 6. Forward selection will result in x_6 being selected with performance 0.18749 and subsequently x_3 with performance 0.17624 and then x_5 with performance 0.17082 as no improvement can be achieved by adding additional features it will terminate at the set x_3, x_5 , and x_6 . Backward selection will result in removing x_4 to have x_2, x_3, x_5, x_6 with performance 0.17299 and then remove x_2 to attain the performance 0.17082 by se the feature set x_3, x_5 , and x_6 upon which no further improvements can be achieved by removing features.

Feature(s)	Training RMSE	Test RMSE
none	0.11279	0.20677
x_2	0.10930	0.22301
x_3	0.10974	0.21773
x_4	0.10911	0.21362
x_5	0.11254	0.20729
x_6	0.09301	0.18749
x_2, x_3	0.10914	0.22247
x_2, x_4	0.10756	0.22145
x_2, x_5	0.10909	0.22513
x_2, x_6	0.09108	0.18555
x_3, x_4	0.10837	0.21768
x_3, x_5	0.10961	0.21800
x_3, x_6	0.09108	0.17624
x_4, x_5	0.10910	0.21368
x_4, x_6	0.09234	0.19121
x_5, x_6	0.08993	0.17657
x_2, x_3, x_4	0.10753	0.22138
x_2, x_3, x_5	0.10887	0.22435
x_2, x_3, x_6	0.09071	0.18029
x_2, x_4, x_5	0.10731	0.22315
x_2, x_4, x_6	0.08947	0.19339
x_2, x_5, x_6	0.08900	0.17610
x_3, x_4, x_5	0.10828	0.21795
x_3, x_4, x_6	0.08805	0.17900
x_3, x_5, x_6	0.08896	0.17082
x_4, x_5, x_6	0.08891	0.18062
x_2, x_3, x_4, x_5	0.10730	0.22314
x_2, x_3, x_4, x_6	0.08782	0.18371
x_2, x_3, x_5, x_6	0.08878	0.17299
x_2, x_4, x_5, x_6	0.08727	0.18336
x_3, x_4, x_5, x_6	0.08603	0.17440
x_2, x_3, x_4, x_5, x_6	0.08595	0.17685

Table 2: Root-mean-square error (RMSE) for the training and test set when using least squares regression to predict $\tilde{y}$ using different combinations of the five attributes (x_2 – x_6).

Question 7. We would again like to predict $\tilde{y}$ based on x_2, x_3, x_4, x_5 , and x_6 . For this purpose, we will use the regularized least squares regression which minimizes with respect to $\mathbf{w}$ the following cost function:
 $E(\mathbf{w}) = \sum_n (\tilde{y}_n - [1 \ x_{n2} \ x_{n3} \ x_{n4} \ x_{n5} \ x_{n6}] \mathbf{w})^2 + \lambda \mathbf{w}^\top \mathbf{w}$,
 We will consider 20 different values of λ and use 10-fold cross-validation to select for the optimal value of λ . Which one of the following statement regarding the described regularized least squares regression procedure is correct?

- A. Increasing λ will result in an increase in the 2-norm of the trained $\mathbf{w}$, i.e. in an increase of the quantity $\|\mathbf{w}\|_2$.
- B. 10-fold cross-validation will require the fitting of 10 models in total to quantify the best value of λ .
- C. The test error obtained for the optimal value of λ is a biased estimate of the generalization error.**
- D. When using regularization in least squares regression the model becomes more prone to overfitting.
- E. Don't know.

Solution 7. When increasing λ the quantity $\mathbf{w}^\top \mathbf{w} = \|\mathbf{w}\|_2^2$ will be penalized more and therefore reduced (and not increased in terms of the quantity $\|\mathbf{w}\|_2$). To quantify the best value of λ using 10-fold cross-validation we need to carry out the cross-validation for each of the 20 considered values of λ resulting in $10 \times 20 = 200$ models to be fitted. The test error obtained for the optimal value of λ is biased as it is the best among 20 selected test-performances. To get an unbiased estimate thus requires two-level cross-validation. Regularization reduce overfitting as the model can less well fit the training data due to the regularization.

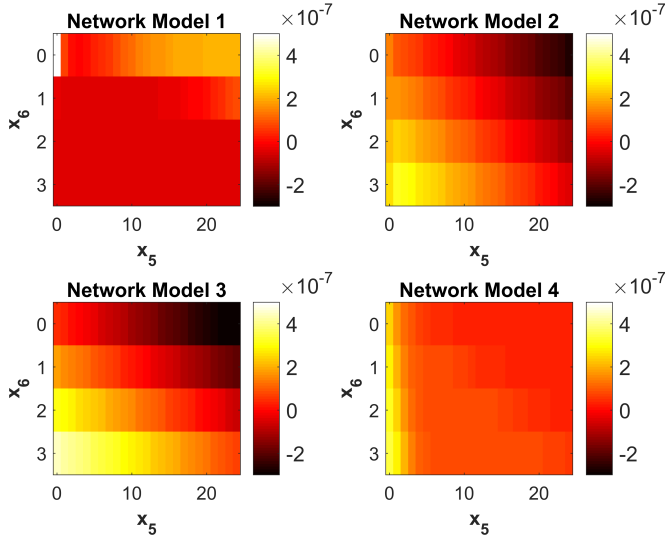


Figure 4: Four different artificial neural network models trained on the airline data to predict $\tilde{y}$ based on the features x_5 and x_6 . In each image plot is given the output of one of the four networks using different combinations of x_5 and x_6 . As such, $x_5 = 0$ and $x_6 = 0$ is given at the upper left corner whereas $x_5 = 24$ and $x_6 = 3$ is given at the lower right corner of each image.

Question 8. We will consider an artificial neural network (ANN) trained to predict $\tilde{y}$ based only on using x_5 and x_6 as inputs, i.e., incidences and fatal accidents during 2000-2014. The trained model is given by $f(\mathbf{x}, \mathbf{w}) = w_0^{(2)} + \sum_{j=1}^2 w_j^{(2)} h^{(1)}([1 \ x_5 \ x_6] \mathbf{w}_j^{(1)})$, where $h^{(1)}(z) = 1/(1 + \exp(-z))$ is the logistic function used as activation function in the hidden layer (i.e., values are mapped to be between 0 and 1). We will consider an ANN with two hidden units in the hidden layer defined by:

$$\mathbf{w}_1^{(1)} = \begin{bmatrix} 0.0189 \\ 0.9159 \\ -0.4256 \end{bmatrix}, \mathbf{w}_2^{(1)} = \begin{bmatrix} 3.7336 \\ -0.8003 \\ 5.0741 \end{bmatrix},$$

and $w_0^{(2)} = 0.3799 \cdot 10^{-6}$, $w_1^{(2)} = -0.3440 \cdot 10^{-6}$, and $w_2^{(2)} = 0.0429 \cdot 10^{-6}$. Which one of the resulting outputs as function of x_5 and x_6 given in Figure 4 corresponds to the trained network?

- A. Network Model 1.
- B. Network Model 2.
- C. Network Model 3.

D. Network Model 4.

- E. Don't know.

Solution 8. Consider for instance the two corners located at $x_5 = 0, x_6 = 3$ and $x_5 = 24, x_6 = 0$ at these two locations we have that the outputs are respectively given by:

$$\begin{aligned} f([03], \mathbf{w}) &= 0.3799 \cdot 10^{-6} \\ &\quad - 0.3440 \cdot 10^{-6} \cdot \frac{1}{1 + \exp(-([1 \ 0 \ 3] \cdot \begin{bmatrix} 0.0189 \\ 0.9159 \\ -0.4256 \end{bmatrix}))} \\ &\quad + 0.0429 \cdot 10^{-6} \cdot \frac{1}{1 + \exp(-([1 \ 0 \ 3] \cdot \begin{bmatrix} 3.7336 \\ -0.8003 \\ 5.0741 \end{bmatrix}))} \\ &= 0.3799 \cdot 10^{-6} - 0.3440 \cdot 10^{-6} \cdot 0.2213 \\ &\quad + 0.0429 \cdot 10^{-6} \cdot 1.0000 \\ &= 3.4667 \cdot 10^{-7} \end{aligned}$$

and

$$\begin{aligned} f([240], \mathbf{w}) &= 0.3799 \cdot 10^{-6} \\ &\quad - 0.3440 \cdot 10^{-6} \cdot \frac{1}{1 + \exp(-([1 \ 24 \ 0] \cdot \begin{bmatrix} 0.0189 \\ 0.9159 \\ -0.4256 \end{bmatrix}))} \\ &\quad + 0.0429 \cdot 10^{-6} \cdot \frac{1}{1 + \exp(-([1 \ 24 \ 0] \cdot \begin{bmatrix} 3.7336 \\ -0.8003 \\ 5.0741 \end{bmatrix}))} \\ &= 0.3799 \cdot 10^{-6} - 0.3440 \cdot 10^{-6} \cdot 1 \\ &\quad + 0.0429 \cdot 10^{-6} \cdot 4.2410e \cdot 10^{-7} \\ &= 3.5900 \cdot 10^{-8}. \end{aligned}$$

The only network that has this property is Network Model 4.

Question 9. We would like to use two level cross-validation to select for the optimal number of hidden units in an artificial neural network (ANN) with one hidden layer as well as quantify the generalization of the selected model. For this purpose, we will use two-level cross-validation in which we in the outer fold use 5-fold cross-validation and in the inner fold (i.e., the fold in which we quantify the optimal number of hidden units in the hidden layer) use 10-fold cross-validation. As ANNs are prone to local minima issues we will train three models for each specification of the number of hidden unit based on three different random initializations and use the model out of these three with best training error. We have a computational budget of training 1000 models and would like to evaluate in steps of 1 from 1 to H hidden units (i.e., if $H=3$ we will evaluate ANNs with 1, 2, and 3 hidden units). What is the largest value of H for which no more than 1000 models will be trained?

A. 6

B. 19

C. 20

D. 66

E. Don't know.

Solution 9. In two level cross-validation we have for each inner fold $10 \cdot H$ different models. For the optimal of these selected models, we will have to train an additional model on the full dataset used for the inner fold to predict the test data of the outer fold. This thus requires the training of $5 \cdot (10 \cdot H + 1)$ models. As we for each trained model use three random initializations we obtain a total of $3 \cdot 5 \cdot (10 \cdot H + 1)$ models to be trained. We thereby obtain $3 \cdot 5 \cdot (10 \cdot H + 1) = 1000 \Rightarrow (10 \cdot H + 1) = 1000/15 \Rightarrow H = (1000/15 - 1)/10 = 6.5667$. We can thus maximally evaluate for $H=6$.

Question 10. Some people are afraid of flying and thus prefer to take for instance the car or bus. According to the economist Ian Savage² the probability of dying travelling 600 km is approximately:

- Chance dying travelling by car is 0.000271 %.
- Chance dying travelling by bus is 0.000004 %.
- Chance dying travelling by plane is 0.000003 %.

The distance travelling from Copenhagen to Oslo is 600 km regardless of the trip being based on car, bus, or plane. We will assume when travelling from Copenhagen to Oslo 30 % of people take the car, 10 % of people take the bus, and 60 % of people take the plane. Given a person died travelling between Copenhagen and Oslo what is the probability it was from travelling by plane?

A. $1.80 \cdot 10^{-4}$ %

B. 1.08 %

C. 2.16 %

D. 10.0 %

E. Don't know.

Solution 10. Let D denote the event dying travelling between Copenhagen and Oslo. Let C denote the event car, B the event Buss and F the event plane (i.e., flight). According to the numbers given we have $P(D|C)=0.000271$ %, $P(D|B)=0.000004$ %, and $P(D|F)=0.000003$ %. Using Bayes theorem we have:

$$\begin{aligned} P(F|D) &= \frac{P(D|F)P(F)}{P(D|F)P(F) + P(D|B)P(B) + P(D|C)P(C)} \\ &= \frac{0.000003\% \cdot 60\%}{0.000003\% \cdot 60\% + 0.000004\% \cdot 10\% + 0.000271\% \cdot 30\%} \\ &= 2.16\% \end{aligned}$$

²<http://faculty.wcas.northwestern.edu/~ipsavage/MosesLecture.pdf>

Question 11. We will predict whether an airline company is relatively safe (considered the positive class) or unsafe (considered the negative class) based on thresholding $\tilde{y}$ at its median value, i.e. if $\tilde{y} < \text{median}(\tilde{y})$ it is considered safe otherwise it is considered unsafe.

A decision tree is subsequently fitted to the data. At the root of the tree it is considered to split according to the median value of the number of incidences in 2000-2014 (i.e., x_5). For impurity we will use the classification error given by $I(v) = 1 - \max_c p(c|v)$. Before the split, we have 32 safe and 24 unsafe airline companies, and after the split we have

- 23 safe and 8 unsafe airline companies with relatively few incidences.
- 9 safe and 16 unsafe airline companies with relatively many incidences.

Which statement regarding the purity gain Δ of the split is correct?

- A. $\Delta = -0.2679$
- B. $\Delta = 0.1250$**
- C. $\Delta = 0.2500$
- D. $\Delta = 0.4286$
- E. Don't know.

Solution 11. The purity gain is given by

$$\Delta = I(r) - \sum_{k=1}^K \frac{N(v_k)}{N} I(v_k),$$

where

$$I(v) = 1 - \max_c p(c|v).$$

Evaluating the purity gain for the split we have:

$$\begin{aligned} \Delta &= (1 - 32/56) \\ &\quad - \left[\frac{31}{56} \left(1 - \left(\frac{23}{31} \right) \right) \right] \\ &\quad + \frac{25}{56} \left(1 - \left(\frac{16}{25} \right) \right) \\ &= 7/56 \end{aligned}$$

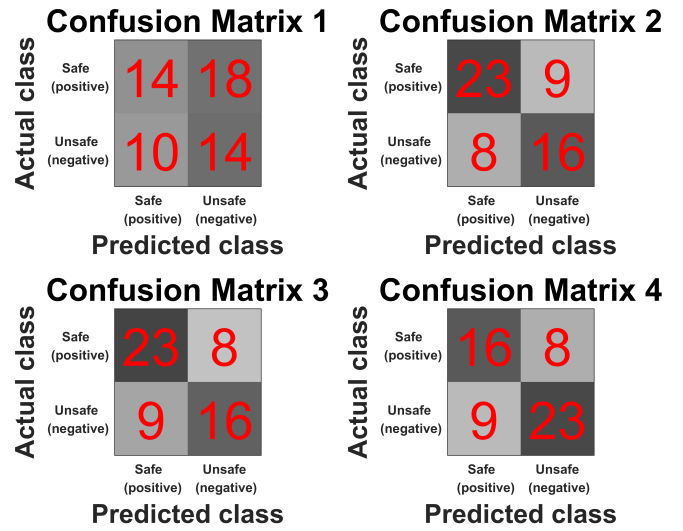


Figure 5: Four different confusion matrices where one corresponds to the confusion matrix of the decision tree with one split according to the median value of the number of incidences in 2000-2014 (i.e., x_5).

Question 12. We will consider the decision tree given by having only the above split (defined in question 11) as a decision and classifying according to this split using the largest class (i.e., using majority voting). In Figure 5 is given four different confusion matrices. Which one of the four confusion matrices corresponds to the decision tree's classification of the 56 observations?

- A. Confusion Matrix 1.
- B. Confusion Matrix 2.**
- C. Confusion Matrix 3.
- D. Confusion Matrix 4.
- E. Don't know.

Solution 12. The decision tree will use majority voting at each leaf in order to classify the 56 observations. For the left branch we have that the majority is safe and thus the 23 safe observations will be correctly classified whereas the 8 unsafe classification will be misclassified as safe. Likewise for the right branch, the majority is unsafe and thus the 16 unsafe observations will be classified as unsafe whereas the 8 safe observations will be misclassified as unsafe. This corresponds to confusion matrix 2.

Question 13. Which statement regarding classification is correct?

- A. In classification the output value is continuous.
- B. Logistic regression is not a classification approach but a regression method.
- C. The k-means algorithm is a supervised classification method.
- D. The softmax function is used to provide the probability that an observation is assigned to each class.**
- E. Don't know.

Solution 13. We make the distinction between a classification and a regression problem based on the property of the output variable being either categorical or continuous. Thus, for continuous outputs we use regression methods and not classification methods. Logistic regression is designed for binary outputs and thus a classification method. The k-means algorithm is used for unsupervised learning and thus not a supervised classification method as it only relies on the input data $\mathbf{X}$ and not on output values $\mathbf{y}$. The softmax function is used in multinomial regression and artificial neural networks for multi-class classification problems in order to provide outputs interpreted as the probability an observation is assigned to each class similar to the role of the logistic function in logistic regression.

Question 14. We will consider Confusion Matrix 1 given in Figure 5 and we regard the class safe as the *positive* class and unsafe as the *negative* class. Which statement regarding a classifier having performance given by the performance indicated by Confusion Matrix 1 is correct?

- A. The classifier's precision is 7/12.**
- B. The classifier's recall is 1/2.
- C. The false positive rate (FPR) of the classifier is 5/14.
- D. The accuracy of the classifier is better than guessing everything to be the largest class.
- E. Don't know.

Solution 14. The precision is $14/(14+10)=7/12$. The recall is $14/(14+18)=7/16$. The FPR is $10/(10+14)=5/12$. Guessing everything to be the

largest class would correspond to guessing everything as unsafe with accuracy of 32/56 whereas the accuracy of the classifier is 28/56.

	O1	O2	O3	O4	O5	O6	O7	O8	O9	O10
O1	0	8.55	0.43	1.25	1.14	3.73	2.72	1.63	1.68	1.28
O2	8.55	0	8.23	8.13	8.49	6.84	8.23	8.28	8.13	7.66
O3	0.43	8.23	0	1.09	1.10	3.55	2.68	1.50	1.52	1.05
O4	1.25	8.13	1.09	0	1.23	3.21	2.17	1.29	1.33	0.56
O5	1.14	8.49	1.10	1.23	0	3.20	2.68	1.56	1.50	1.28
O6	3.73	6.84	3.55	3.21	3.20	0	2.98	2.66	2.50	3.00
O7	2.72	8.23	2.68	2.17	2.68	2.98	0	2.28	2.30	2.31
O8	1.63	8.28	1.50	1.29	1.56	2.66	2.28	0	0.25	1.46
O9	1.68	8.13	1.52	1.33	1.50	2.50	2.30	0.25	0	1.44
O10	1.28	7.66	1.05	0.56	1.28	3.00	2.31	1.46	1.44	0

Table 3: Pairwise Euclidean distances between the first ten observations of the standardized airline safety data. Black observations (i.e., O1, O3, O4, O5, O10) are observations corresponding to relatively safe airline companies, red observations (i.e., O2, O6, O7, O8, O9) are observations corresponding to relatively unsafe airline companies.

Question 15. To determine whether an airline company is relatively safe or unsafe we will use a k-nearest neighbor (KNN) classifier to predict each of the ten observations based on the Euclidean distances between the observations given in Table 3. We will use leave-one-out cross-validation for the KNN in order to classify the ten considered observations and use $K = 1$, i.e., a one nearest neighbor classifier. The analysis will be based only on the data given in Table 3. What will be the error rate of the classifier?

- A. 0 %
- B. 10 %**
- C. 20 %
- D. 30 %
- E. Don't know.

Solution 15. $N(O1, 1) = \{O3\}$ as O3 is closest it will be correctly classified as safe.

$N(O2, 1) = \{O6\}$ as O6 is closest it will be correctly classified as unsafe.

$N(O3, 1) = \{O1\}$ as O1 is closest it will be correctly classified as safe.

$N(O4, 1) = \{O10\}$ as O10 is closest it will be correctly classified as safe.

$N(O5, 1) = \{O3\}$ as O3 is closest it will be correctly classified as safe.

$N(O6, 1) = \{O9\}$ as O9 is closest it will be correctly classified as unsafe.

$N(O7, 1) = \{O4\}$ as O4 is closest it will be incorrectly classified as safe.

$N(O8, 1) = \{O9\}$ as O9 is closest it will be correctly classified as unsafe.

$N(O9, 1) = \{O8\}$ as O8 is closest it will be correctly classified as unsafe.

$N(O10, 1) = \{O4\}$ as O4 is closest it will be correctly classified as safe.

Thus, one out of the ten observations will be misclassified.

Question 16. We will again consider the Euclidean distances between the first ten observations given in Table 3. Agglomerative hierarchical clustering is used to cluster these ten observations based on their distances to each other using average linkage. Which one of the dendrograms given in Figure 6 corresponds to the clustering?

- A. Dendrogram 1.
- B. Dendrogram 2.
- C. Dendrogram 3.
- D. Dendrogram 4.**
- E. Don't know.

Solution 16. As O2 merges last to the cluster containing all the remaining observations we can simply evaluate at what level O2 will merge which is given by O2's average distance to the observations $\{O1, O3, O4, O5, O6, O7, O8, O9, O10\}$ which is given by $(8.55 + 8.23 + 8.13 + 8.49 + 6.84 + 8.24 + 8.28 + 8.13 + 7.66)/9 = 8.0611$. Only dendrogram 4 has this property.

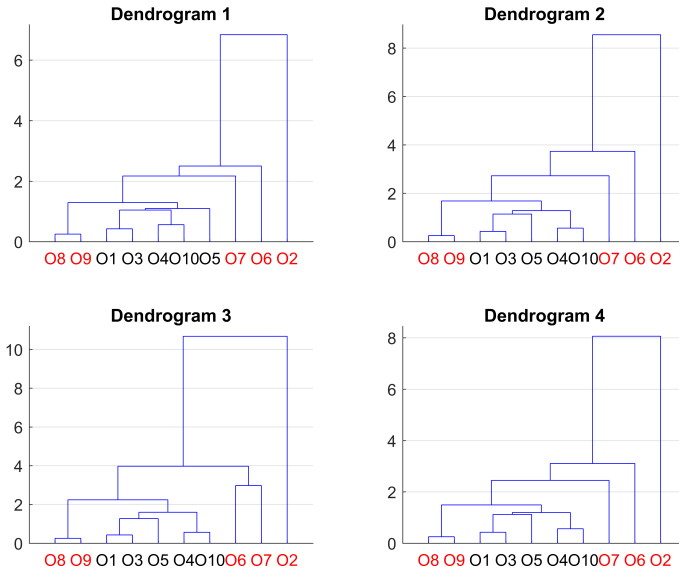


Figure 6: Four different dendrograms derived using the Euclidean distances between the first 10 observations in the airline safety data. Black observations correspond to relatively safe companies whereas red to relatively unsafe companies.

Question 17. We will cut dendrogram 1 at the level of three clusters and evaluate this clustering in terms of its correspondence with the class label information in which black observations, i.e., O1, O3, O4, O5, O10, are observations corresponding to relatively safe airline companies, and red observations, i.e., O2, O6, O7, O8, O9, are observations corresponding to relatively unsafe airline companies. We recall that the Rand index also denoted the simple matching coefficient (SMC) between the true labels and the extracted clusters is given by $R = \frac{f_{11} + f_{00}}{K}$, where f_{11} is the number of object pairs in same class assigned to same cluster, f_{00} is the number of object pairs in different class assigned to different clusters, and $K = N(N - 1)/2$ is the total number of object pairs, where N is the number of observations considered. What is the value of R between the true labeling of the observations and the three extracted clusters?

- A. 0.40
- B. 0.47
- C. 0.51**
- D. 0.60
- E. Don't know.

Solution 17. The cluster indices are given by the vector: $[1211131111]^T$, whereas the true class labels are given by the vector $[1211122221]^T$. From this, we obtain:

$$K = 10(10 - 1)/2 = 45$$

$$f_{00} = 5 \cdot 1 + 5 \cdot 1 + 1 \cdot 0 = 10$$

$$f_{11} = 5 \cdot (5 - 1)/2 + 3 \cdot (3 - 1)/2 + 1 \cdot (1 - 1)/2 + 1 \cdot (1 - 1)/2 = 13$$

$$R = \frac{f_{11} + f_{00}}{K} = \frac{13 + 10}{45} = 23/45.$$

Question 18. We suspect that observation O2 may be an outlier. In order to quantify if this could be the case we will calculate the average relative KNN density based on Euclidean distance and the observations given in Table 3 only. We recall that the KNN density and average relative density (ard) for the observation $\mathbf{x}_i$ are given by:

$$\text{density}_{\mathbf{X}_{\setminus i}}(\mathbf{x}_i, K) = \frac{1}{\frac{1}{K} \sum_{\mathbf{x}' \in N_{\mathbf{X}_{\setminus i}}(\mathbf{x}_i, K)} d(\mathbf{x}_i, \mathbf{x}')},$$

$$\text{ard}_{\mathbf{X}}(\mathbf{x}_i, K) = \frac{\text{density}_{\mathbf{X}_{\setminus i}}(\mathbf{x}_i, K)}{\frac{1}{K} \sum_{\mathbf{x}_j \in N_{\mathbf{X}_{\setminus i}}(\mathbf{x}_i, K)} \text{density}_{\mathbf{X}_{\setminus j}}(\mathbf{x}_j, K)},$$

where $N_{\mathbf{X}_{\setminus i}}(\mathbf{x}_i, K)$ is the set of K nearest neighbors of observation $\mathbf{x}_i$ excluding the i 'th observation, and $\text{ard}_{\mathbf{X}}(\mathbf{x}_i, K)$ is the average relative density of $\mathbf{x}_i$ using K nearest neighbors. What is the average relative density for observation O2 for $K = 2$ nearest neighbors?

- A. 0.085
- B. 0.138
- C. 0.169**
- D. 0.356
- E. Don't know.

Solution 18.

$$\begin{aligned} \text{density}(\mathbf{x}_{O2}, 2) &= \left(\frac{1}{2}(6.84 + 7.66)\right)^{-1} = 0.1379 \\ \text{density}(\mathbf{x}_{O6}, 2) &= \left(\frac{1}{2}(2.50 + 2.66)\right)^{-1} = 0.3876 \\ \text{density}(\mathbf{x}_{O10}, 2) &= \left(\frac{1}{2}(0.56 + 1.05)\right)^{-1} = 1.2422 \\ \text{a.r.d.}(\mathbf{x}_{O2}, 2) &= \frac{\text{density}(\mathbf{x}_{O2}, 2)}{\frac{1}{2}(\text{density}(\mathbf{x}_{O6}, 2) + \text{density}(\mathbf{x}_{O10}, 2))} \\ &= \frac{0.1379}{\frac{1}{2}(0.3876 + 1.2422)} = 0.169 \end{aligned}$$

	x_1^L	x_1^H	x_2^L	x_2^H	x_3^L	x_3^H	x_4^L	x_4^H	x_5^L	x_5^H	x_6^L	x_6^H
O1	1	0	1	0	1	0	1	0	1	0	1	0
O2	0	1	0	1	0	1	0	1	0	1	0	1
O3	1	0	0	1	1	0	1	0	1	0	1	0
O4	1	0	1	0	1	0	0	1	0	1	1	0
O5	0	1	1	0	1	0	1	0	1	0	1	0
O6	0	1	0	1	0	1	0	1	0	1	0	1
O7	0	1	1	0	1	0	0	1	0	1	0	1
O8	1	0	1	0	1	0	1	0	0	1	0	1
O9	0	1	0	1	1	0	1	0	0	1	0	1
O10	1	0	0	1	0	1	0	1	0	1	1	0

Table 4: The ten first observations of the airline safety dataset binarized considering the attribute x_1 – x_6 . The attributes are all binarized according to being below or equal (denoted L) or above the median value (denoted H). The ten observations are color coded in terms of relatively safe {O1, O3, O4, O5, O10} or unsafe {O2, O6, O7, O8, O9} airline companies.

Question 19. We will binarize each feature in the airline safety data according to the median value of the feature denoting below or equal the median value using the superscript L and above the median value using the superscript H . In Table 4 is given the first 10 observations after this binarization and we will consider these 10 observations as a dataset used for market basket analysis with observation O1–O10 corresponding to customers. What is the support for the association rule $\{x_2^H, x_3^H, x_4^H, x_5^H\} \rightarrow \{x_6^H\}$?

- A. 0.0 %
- B. 20.0 %**
- C. 66.7 %
- D. 100.0 %
- E. Don't know.

Solution 19. The support of $\{x_2^H, x_3^H, x_4^H, x_5^H\} \rightarrow \{x_6^H\}$ is given by the number of times out of the total number of customers that customers have relatively high values of both x_2, x_3, x_4, x_5 , and x_6 , i.e., given by the support of the itemset $\{x_2^H, x_3^H, x_4^H, x_5^H, x_6^H\}$. Only customer O2 and O6 have this property out of the 10 customers, thus the support is 2/10.

Question 20. We consider again the data in Table 4 as a market basket problem. What is the confidence of the association rule $\{x_2^H, x_3^H, x_4^H, x_5^H\} \rightarrow \{x_6^H\}$?

- A. 0.0 %
- B. 20.0 %
- C. 66.7 %**
- D. 100.0 %
- E. Don't know.

Solution 20. The confidence is given as

$$\begin{aligned} P(x_6^H = 1 | x_2^H = 1, x_3^H = 1, x_4^H = 1, x_5^H = 1) &= \\ \frac{P(x_6^H = 1, x_2^H = 1, x_3^H = 1, x_4^H = 1, x_5^H = 1)}{P(x_2^H = 1, x_3^H = 1, x_4^H = 1, x_5^H = 1)} &= \\ = \frac{2/10}{3/10} = 2/3 = 66.7\% \end{aligned}$$

Question 21. We would like to predict whether an airline company is relatively safe or unsafe considering only the data given in Table 4. We will apply a Naïve Bayes classifier that assumes independence between the attributes given the class label (i.e., the class label is given by safe airlines in black, i.e., O1, O3, O4, O5, and O10, and unsafe airlines in red, i.e., O2, O6, O7, O8, and O9). Given that an airline company has $x_2^H = 1$, $x_3^H = 1$, $x_4^H = 1$, $x_5^H = 1$ what is the probability that the airline company is considered safe according to the Naïve Bayes classifier derived from the data in Table 4?

- A. 0
- B. 4/625
- C. 4/49**
- D. 1/3
- E. Don't know.

Solution 21. Let $\tilde{y} = 1$ denote that the airline is safe.

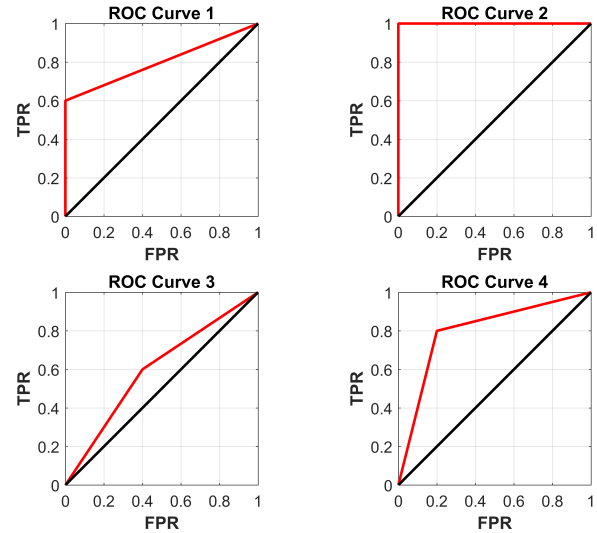


Figure 7: Four different receiver operator characteristic (ROC) curves.

According to the Naïve Bayes classifier we have

$$\begin{aligned} P(\tilde{y} = 1 | x_2^H = 1, x_3^H = 1, x_4^H = 1, x_5^H = 1) &= \\ \frac{\begin{pmatrix} P(x_2^H = 1 | \tilde{y} = 1) \times \\ P(x_3^H = 1 | \tilde{y} = 1) \times \\ P(x_4^H = 1 | \tilde{y} = 1) \times \\ P(x_5^H = 1 | \tilde{y} = 1) \times \\ P(\tilde{y} = 1) \end{pmatrix}}{\begin{pmatrix} P(x_2^H = 1 | \tilde{y} = 1) \times \\ P(x_3^H = 1 | \tilde{y} = 1) \times \\ P(x_4^H = 1 | \tilde{y} = 1) \times \\ P(x_5^H = 1 | \tilde{y} = 1) \times \\ P(\tilde{y} = 1) \\ + P(x_2^H = 1 | \tilde{y} = 0) \times \\ P(x_3^H = 1 | \tilde{y} = 0) \times \\ P(x_4^H = 1 | \tilde{y} = 0) \times \\ P(x_5^H = 1 | \tilde{y} = 0) \times \\ P(\tilde{y} = 0) \end{pmatrix}} &= \\ = \frac{2/5 \cdot 1/5 \cdot 2/5 \cdot 2/5 \cdot 5/10}{2/5 \cdot 1/5 \cdot 2/5 \cdot 2/5 \cdot 5/10 + 3/5 \cdot 2/5 \cdot 3/5 \cdot 5/5 \cdot 5/10} &= \\ = \frac{40/(5^4 \cdot 10)}{40/(5^4 \cdot 10) + 450/(5^4 \cdot 10)} = 40/490 = 4/49 \end{aligned}$$

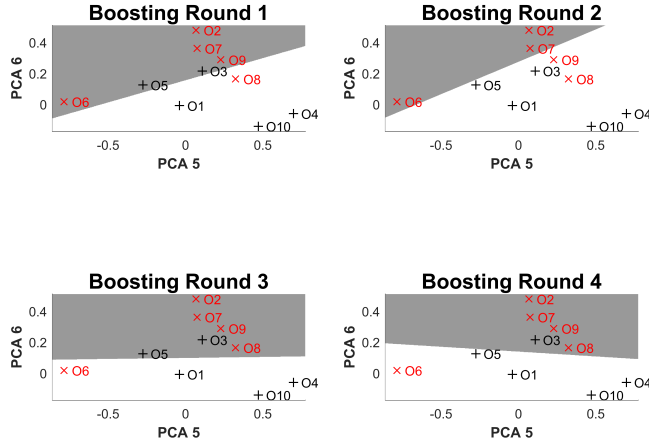


Figure 8: Decision boundaries for four rounds of boosting considering a logistic regression model using the fifth and sixth principal components as features and the first 10 observations of the airline safety data. Gray region indicates that the observation will be classified as unsafe (red crosses), white regions that the observation will be classified as safe (black plusses).

	Round 1	Round 2	Round 3	Round 4
O1	0.1000	0.0714	0.0469	0.0319
O2	0.1000	0.0714	0.0469	0.0319
O3	0.1000	0.1667	0.1094	0.2059
O4	0.1000	0.0714	0.0469	0.0319
O5	0.1000	0.1667	0.1094	0.2059
O6	0.1000	0.0714	0.0469	0.0882
O7	0.1000	0.0714	0.0469	0.0319
O8	0.1000	0.1667	0.3500	0.2383
O9	0.1000	0.0714	0.1500	0.1021
O10	0.1000	0.0714	0.0469	0.0319

Table 5: The weights for the first four rounds of AdaBoost.

Question 22. We will use x_5^L to determine if an airline is safe *considered the positive class* or unsafe *considered the negative class*. In Figure 7 are given four different receiver operator characteristic curves (ROC). Which one of the four ROC curves corresponds to using x_5^L to determine if an airline is safe (positive class) or unsafe (negative class)?

A. ROC curve 1.

B. ROC curve 2.

C. ROC curve 3.

D. ROC curve 4.

E. Don't know.

Solution 22. The ROC curve starts at (0,0) for which we threshold above 1. When tresholding at one we obtain that 3 out of 5 safe airlines (positive class) have $x_5^L = 1$ and 0 out of 5 unsafe (negative class) have $x_5^L = 1$ thus the ROC curve will be at the point (0,3/5). When we subsequently further lower the threshold to be at 0 all airline companies that are safe and unsafe will be at this threshold value or above, thus, the ROC curve will end here at (1,1). Only ROC curve 1 has this property.

Question 23. We would like to build a model for classifying whether an airline is safe or not based on the first 10 observations of the airline safety dataset. In order to do so, we will use the resulting classifier obtained by using the AdaBoost algorithm and a logistic regression classifier. The weights of the first four rounds of the AdaBoost procedure is given in Figure 8 and the associated sampling weights used for each round is given in Table 5. How will observation O5 and O6 be classified according to the ensemble classifier obtained by combining the four boosting rounds using the voting procedure defined by the AdaBoost algorithm?

- A. Observation O5 and O6 will be tied between safe and unsafe by the AdaBoost classifier.
- B. Both observation O5 and O6 will be correctly classified by the AdaBoost classifier.**
- C. Only one of the two observations O5 and O6 will be correctly classified by the AdaBoost classifier.
- D. Neither of the two observations O5 and O6 will be correctly classified by the AdaBoost classifier.
- E. Don't know.

Solution 23. The resulting boosting procedure weight each classifier according to their importance α_t for the t^{th} round, where $\alpha_t = 0.5 \log \frac{1-e_t}{e_t}$ such that $e_t = \sum_n w_n(t)(1 - \delta_{f_t(x_n), y_n})$ is the error rate weighted according to the weights of the t^{th} round. During the first round O3, O5 and O8 are misclassified and thus $e_1 = 0.1 + 0.1 + 0.1 = 0.3$ and consequently $\alpha_1 = 0.5 \log \frac{1-0.3}{0.3} = 0.4236$. In the second round O8 and O9 are mis-classified and thus $e_2 = 0.1667 + 0.0714 = 0.2381$ and therefore $\alpha_2 = 0.5 \log \frac{1-0.2381}{0.2381} = 0.5816$. In the third round O3, O5, and O6 are misclassified thus $e_3 = 0.1094 + 0.1094 + 0.0469 = 0.2657$ and therefore $\alpha_3 = 0.5 \log \frac{1-0.2657}{0.2657} = 0.5083$. Finally, in the fourth round O3 and O6 are misclassified and thus $e_4 = 0.2059 + 0.0882 = 0.2941$ and therefore $\alpha_4 = 0.5 \log \frac{1-0.2941}{0.2941} = 0.4378$. For observation O5 both classifier of round 2 and 4 vote for it being safe with the strength of vote given by $\alpha_2 + \alpha_4 = 0.5816 + 0.4378 = 1.0194$ whereas the strength of vote for unsafe is lower, i.e. $\alpha_1 + \alpha_3 = 0.4236 + 0.5083 = 0.9319$. Observation O6 will be classified as unsafe as classifier 1 and 2 classifies it as unsafe with strength $\alpha_1 + \alpha_2 = 0.4236 + 0.5816 = 1.0052$ and safe according

to classifier 3 and 4 with the lower voting strength of $\alpha_3 + \alpha_4 = 0.5083 + 0.4378 = 0.9461$.

Question 24. Four different classifiers are trained on the airline safety data considering only the first 10 observations projected onto the fifth and sixth principal component to determine if an airline is relatively safe or unsafe. The decision boundary for each of the four classifiers is given in Figure 9 when only using as input the data projected onto the fifth (PCA 5) and sixth (PCA 6) principal component, i.e. each method has only these two inputs. Which one of the following statements is correct?

- A. Classifier 1 corresponds to a logistic regression classifier, Classifier 2 is a 3-nearest neighbor classifier using Euclidean distance, Classifier 3 is a decision tree classifier, and Classifier 4 corresponds to an artificial neural network (ANN).
- B. Classifier 1 is a Naive Bayes classifier based on the use of univariate normal distributions, Classifier 2 is a 3-nearest neighbor classifier, Classifier 3 is a Decision Tree classifier, and Classifier 4 is a 1-nearest neighbor classifier using Euclidean distance.
- C. Classifier 1 is a Naive Bayes classifier based on the use of univariate normal distributions, Classifier 2 is a 3-nearest neighbor classifier, Classifier 3 is a 1-nearest neighbor classifier, and Classifier 4 is a Decision Tree classifier.
- D. Classifier 1 is a 3-nearest neighbor classifier using Euclidean distance, Classifier 2 is a Naive Bayes classifier based on the use of univariate normal distributions, Classifier 3 is a Decision Tree classifier, and Classifier 4 is a 1-nearest neighbor classifier using Euclidean distance.**
- E. Don't know.

Solution 24. Classifier 1 is a 3-nearest neighbor classifier as such one red cross and one black plus are within the wrong decision boundary due to the majority voting of the three nearest neighbors. Classifier 2 has smooth decision boundaries which would correspond to the Naive Bayes classifier based on the use of univariate normal distributions. Classifier 3 is a Decision Tree classifier due to its vertical and horizontal decision boundaries. Classifier 4 is a 1-nearest neighbor classifier using Euclidean distance as the decision boundary clearly follows the most nearby observation.

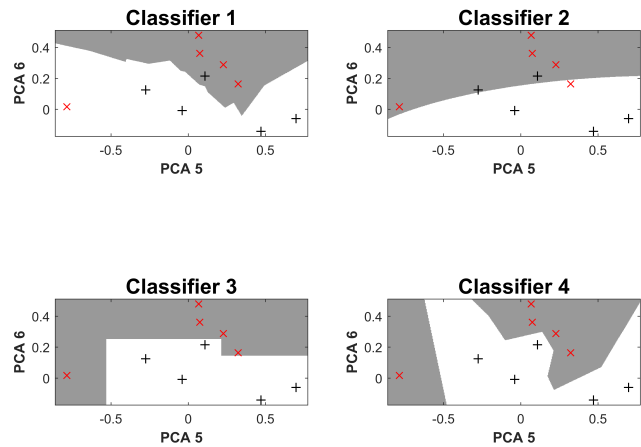


Figure 9: Decision boundaries for four different classifiers trained on the airline safety data using principal component 5 and 6 as input to the classifiers. Gray regions classify into red crosses whereas white regions into black plusses.

Question 25. Consider a dataset with eight observations located at $\{1.0, 1.2, 1.5, 2.0, 2.2, 2.5, 3.0, 3.2\}$. We will cluster the dataset using the k-means algorithm using $k = 3$ clusters and initialize the clusters at the locations of the first three observations, i.e. cluster 1 will be initially located at 1.0, cluster 2 at 1.2 and cluster 3 at 1.5. What will be the converged clustering of the eight observations using the k-means procedure based on Euclidean distance as dissimilarity?

- A. $\{1.0\}, \{1.2, 1.5\}, \{2.0, 2.2, 2.5, 3.0, 3.2\}$**
- B. $\{1.0, 1.2\}, \{1.5\}, \{2.0, 2.2, 2.5, 3.0, 3.2\}$
- C. $\{1.0, 1.2, 1.5\}, \{2.0, 2.2, 2.5\}, \{3.0, 3.2\}$
- D. $\{1.0, 1.2, 1.5\}, \{2.0, 2.2\}, \{2.5, 3.0, 3.2\}$
- E. Don't know.

Solution 25. The cluster located at 1.5 will be closest to the observations located at 2.0, 2.2, 2.5, 3.0, and 3.2 and will therefore be assigned these whereas cluster located at 1.0 and 1.2 will only be assigned respectively the observation located at 1.0 and 1.2. The location of the centroid for cluster 3 will thereby be changed such that cluster 3 is updated to be located at: $(1.5 + 2.0 + 2.2 + 2.5 + 3.0 + 3.2)/6 = 2.4$. Subsequently, observation 1.5 will be closer to cluster located at 1.2 than cluster located at 2.4 and will therefore as only observation change assignment, such that cluster 2 will

be updated to be located at $(1.2 + 1.5)/2 = 1.35$ whereas cluster 3 will be updated to be located at $(2.0 + 2.2 + 2.5 + 3.0 + 3.2)/5 = 2.58$. As no observation will change assignment based on the location of the updated clusters the algorithm will converge to the clustering given by:
 $\{1.0\}$, $\{1.2, 1.5\}$, $\{2.0, 2.2, 2.5, 3.0, 3.2\}$.

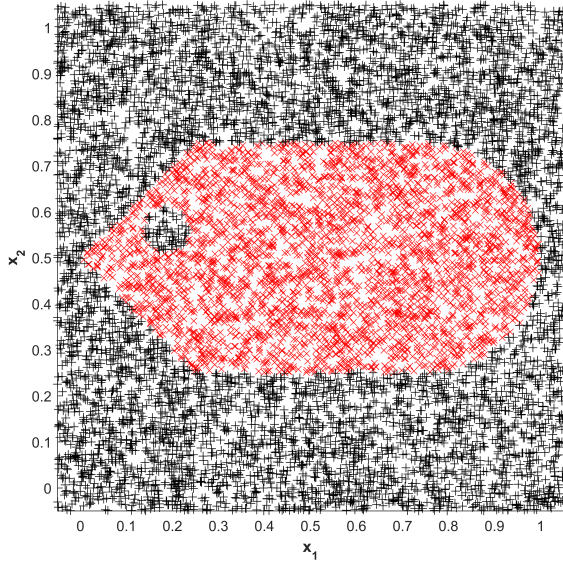


Figure 10: A two class classification problem with red crosses (i.e., $\times$) and black pluses (i.e., $+$) constituting the two classes.

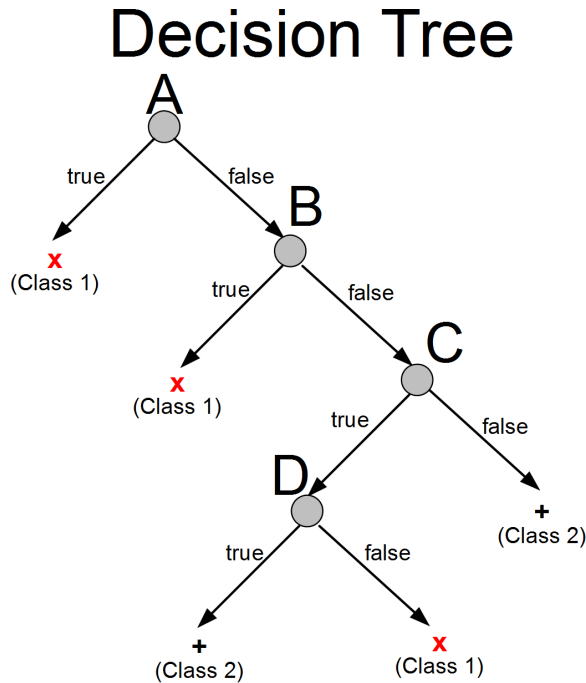


Figure 11: A decision tree with four decisions (A, B, C, and D) perfectly separating the black pluses from red crosses in Figure 10 if adequately defined.

Question 26. We will consider the two class classification problem given in Figure 10 in which the goal is to separate red crosses (i.e., $\times$) from black pluses (i.e., $+$). Which one of the following procedures based on the decision tree given in Figure 11 will perfectly separate the two classes?

A. $A = \|x - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}\|_1 \leq 1/4,$
 $B = \|x - \begin{bmatrix} 3/4 \\ 1/2 \end{bmatrix}\|_2 \leq 1/20,$
 $C = \|x - \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}\|_\infty \leq 1/4,$
 $D = \|x - \begin{bmatrix} 3/16 \\ 9/16 \end{bmatrix}\|_2 \leq 1/4.$

B. $A = \|x - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}\|_1 \leq 1/4,$
 $B = \|x - \begin{bmatrix} 3/4 \\ 1/2 \end{bmatrix}\|_2 \leq 1/4,$
 $C = \|x - \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}\|_\infty \leq 1/4,$
 $D = \|x - \begin{bmatrix} 3/16 \\ 9/16 \end{bmatrix}\|_2 \leq 1/20.$

C. $A = \|x - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}\|_\infty \leq 1/4,$
 $B = \|x - \begin{bmatrix} 3/4 \\ 1/2 \end{bmatrix}\|_1 \leq 1/4,$
 $C = \|x - \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}\|_2 \leq 1/4,$
 $D = \|x - \begin{bmatrix} 3/16 \\ 9/16 \end{bmatrix}\|_2 \leq 1/20.$

D. $A = \|x - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}\|_\infty \leq 1/4,$
 $B = \|x - \begin{bmatrix} 3/4 \\ 1/2 \end{bmatrix}\|_2 \leq 1/4,$
 $C = \|x - \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}\|_1 \leq 1/4,$
 $D = \|x - \begin{bmatrix} 3/16 \\ 9/16 \end{bmatrix}\|_2 \leq 1/20.$

E. Don't know.

Solution 26. The red crosses contains a circle located at (0.75,0.5) and a square located at (0.5, 0.5) and a diamond located at (0.25, 0.5) all with radius 0.25 corresponding to:

$$A = \|x - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}\|_\infty \leq 1/4,$$

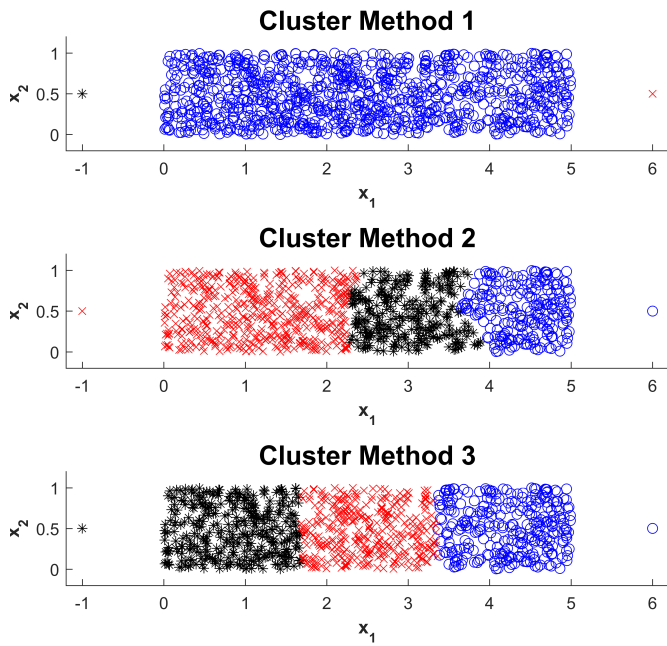


Figure 12: A dataset with two features x_1 and x_2 and 100 observations clustered using three different clustering approaches.

$$B = \left\| \mathbf{x} - \begin{bmatrix} 3/4 \\ 1/2 \end{bmatrix} \right\|_2 \leq 1/4,$$

$$C = \left\| \mathbf{x} - \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix} \right\|_1 \leq 1/4.$$

Finally, there are black plusses located at (0.1875, 0.5625) with a small radius of 0.05 corresponding to:

$$D = \left\| \mathbf{x} - \begin{bmatrix} 3/16 \\ 9/16 \end{bmatrix} \right\|_2 \leq 1/20.$$

This only holds for the last answer option.

Question 27. We will consider a dataset with two features x_1 and x_2 and 1000 observations that is clustered using three different clustering approaches all based on Euclidean distance as measure of distance. The clustering extracted by each of the three considered approaches are given in Figure 12. Which one of the following statements is correct?

- A. Cluster Method 1 corresponds to k-means,
Cluster Method 2 corresponds to hierarchical clustering using single linkage,
Cluster method 3 corresponds to hierarchical clustering using complete linkage.
- B. Cluster Method 1 corresponds to hierarchical clustering using single linkage,
Cluster Method 2 corresponds to k-means,
Cluster method 3 corresponds to hierarchical clustering using complete linkage.
- C. Cluster Method 1 corresponds to hierarchical clustering using single linkage,
Cluster method 2 corresponds to hierarchical clustering using complete linkage,
Cluster Method 3 corresponds to k-means.**
- D. Cluster Method 1 corresponds to hierarchical clustering using complete linkage,
Cluster method 2 corresponds to hierarchical clustering using single linkage,
Cluster Method 3 corresponds to k-means.

E. Don't know.

Solution 27. Single linkage clusters consecutively by merging according to the two observations of each cluster that is closest to each other. As such, the clustering will be influenced by the gaps between the observations and therefore the two outlying observations at (-1,0.5) and (6,0.5) will be merged the latest in the dendrogram resulting in the three clusters given by Cluster Method 1. Complete linkage clusters consecutively by merging according to the two observations of each cluster that is the furthest apart. As such, the clustering will be influenced by how far the cluster extends as well as influenced by earlier merge decisions corresponding to the clustering given by Cluster Method 2. K-means will cluster observations according to their proximity to the center of the cluster which corresponds to a clustering given by Cluster Method 3. Neither Cluster Method 1

and Cluster Method 2 can be k-means as the distance to the centroid of each cluster would result in a different clustering configuration than the ones obtained. Furthermore, cluster method 2 cannot be single linkage as the major gaps to $(-1, 0.5)$ and $(6, 0.5)$ will make these merge latest in the dendrogram. Thus the only correct answer option is:

Cluster Method 1 corresponds to hierarchical clustering using single linkage,

Cluster method 2 corresponds to hierarchical clustering using complete linkage,

Cluster Method 3 corresponds to k-means.